Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Ouellette, Amy L Ph.D.		
Project Title:	Development of a Noninvasive, Handheld Immunosensor		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	11/01/2005	**End Date:**	06/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kovacs, Gregory (MENTOR/Stanford University School of Medicine)		
Grant/Contract No.:	NCC 9-58-PF00901		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	POSTDOCTORAL FELLOWSHIP Cytokine proteins are one of the most important early indicators of infection and immune response, and therefore serve as an excellent diagnostic tool for monitoring health. The objective of this project is to develop an easy-to-use handheld diagnostic sensor, with the NSBRI Smart Medical Systems Team, for the rapid detection of several cytokine levels in saliva. This lightweight and non-invasive monitor for the routine detection of cytokine levels would play an important role in astronaut health-care during long-duration space missions, and would be an immediate asset to the medical community on Earth. Current affinity-based technologies used to detect cytokines, such as the enzyme-linked immunosorbent assay (ELISA), are time consuming and labor intensive, requiring trained technicians and bulky equipment not suitable for space flight. The specific aim of this project is to develop a miniature immunosensor that will deliver fast and sensitive results		

Task Description:	(within minutes). This prototype will contain a microfluidic consumable integrated with automated sample preparation and a reusable optical readout device. I will accomplish this aim by optimizing the sensor's three state-of-the-art key features: 1) Adaptation of an affinity-based assay that uses up-converting phosphor (UCP) reporter technology (SRI International, USA) for the sensitive detection and quantification of several cytokine profiles [1]; 2) Modification of micro-pillared capillary structures on the microfluidic chip (Åmic, Sweden) to provide direct fluidic control over critical assay parameters [2]; 3) Development of an optical detector, to detect and amplify UCP signal on the microfluidic chip. This research will produce a convenient and affordable diagnostic test that could also be used to implement virtually any affinity-based assay on a common platform for Earth and space-based medical care.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Earth-based applications of this funded research project are the development of diagnostic equipment to monitor infection and immune response in clinical point-of-care situations. UCP technology integrated with an Åmic microfluidic platform promises a smaller, faster, easy-to-use biosensor that will integrate multiple steps into a single, streamlined application to detect multiple analytes in parallel, that is simple and robust. This immunosensor will have important implications for human healthcare here on Earth. The sensor will also be adaptable to other clinical diagnostic applications for monitoring health and disease, such as cardiac enzymes and pathogen biomarkers.
Task Progress:	The progress of the project's research tasks from November 1, 2005 to June 30, 2006, the main findings are listed below: The first task of the project was to develop a reliable immunoassay using recombinant IFN-y cytokine protein. This was accomplished using two types of assay platforms, a nitrocellulose membrane and a microcapillary chip. Currently in progress, the second task was to implement an automated sample preparation into the microfluidic chip. The third task was to develop and assemble a handheld read-out device to read the immunoassay. Preliminary design and product specifications for this optical detector read-out device were proposed.
Bibliography Type:	Description: (Last Updated: 09/06/2019)
Articles in Peer-reviewed Journals	Ouellette AL, Li JJ, Cooper DE, Ricco AJ, Kovacs GT. "Evolving point-of-care diagnostics using up-converting phosphor bioanalytical systems." Anal Chem. 2009 May 1;81(9):3216-21. https://doi.org/10.1021/ac900475u ; PubMed PMID: 19368339 [NOTE reported originally in January 2007 as "submitted"] , May-2009